INCH-POUND

MIL-DTL-17/234
w/AMENDMENT 2
30 July 2024
MIL-DTL-17/234
w/AMENDMENT 1
09 May 2019

DETAIL SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 75 OHMS, LOW NOISE, LOW SMOKE, ENHANCED INSULATION RESISTANCE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

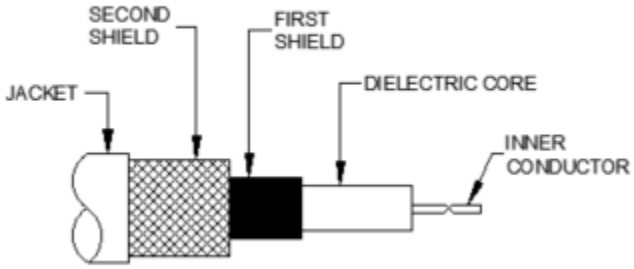


FIGURE 1. General configuration.

TABLE I. Description.

Components	Construction details
Inner Conductor	26 AWG 19x38 silver copper plated alloy Overall diameter: .0189 nominal
Dielectric	Expanded PTFE/PTFE tape composite Diameter: .080 inch ± .004.
First shield	.0015 inch Silver plated copper flat braid having 95% minimum coverage. .086 inch nominal diameter.
Second shield	38 AWG silver plated copper braid having 95% minimum coverage. .104 inch nominal diameter.
Jacket	Extruded; fluorinated ethylene propylene (FEP) Diameter: .125 inch ± .005.

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ENGINEERING INFORMATION:

Continuous working voltage: 3,700 V rms, maximum.

Operating frequency: 1 GHz, maximum.

Velocity of propagation: 82 percent, nominal.

Operating temperature range: -55° to +200°C.

Inner conductor properties:

DC resistance, maximum, at 20°C: 0.65 ohms per 100 feet.

Elongation: 15 percent, minimum.

Tensile strength: 60 $klb_f/inch^2$, minimum.

REQUIREMENTS:

Dimensions, configuration and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination: Applicable.

Diameter: Applicable.

Out-of-roundness: Not Applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors: Inner conductor to core: 7 pounds, minimum; 50 pounds maximum.

Electrical and mechanical:

Continuity: Applicable.

Spark test: 5,000 Vrms, +25 percent, -0 percent.

Voltage withstanding: 10,000 Vrms, minimum.

Insulation resistance:

300,000 megohms, minimum per 1,000 feet (center conductor to outer conductor) at 1,000 V dc.
1,500,000 megohms, minimum per 200 feet (center conductor to outer conductor) at 1,000 V dc.
10 megohms, minimum per 1,000 feet (outer conductor to magnetic shield) at 500 V dc.
50 megohms, minimum per 200 feet (outer conductor to magnetic shield) at 500 V dc.
15 megohms, minimum per 1,000 feet (magnetic shield to outer jacket surface) at 500 V dc.
75 megohms, minimum per 200 feet (magnetic shield to outer jacket surface) at 500V dc.

Corona extinction voltage: 5,000 V rms, minimum.

Characteristic impedance: 75 ±5 ohms.

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RF transmission loss (attenuation): 100MHz - 5dB/100ft
400MHz - 10dB/100ft
500MHz - 12dB/100ft
1GHz - 17dB/100ft
2GHz - 24dB/100ft

Standing wave ratio (return loss): Not applicable.

Capacitance: 17pF per foot, maximum.

Capacitance stability: Not applicable
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Inductance: Not applicable.

Mechanically induced noise voltage: 320 microvolts peak to peak, maximum.

Time delay: Not applicable.

Aging stability: +98° ±2°C.

Stress crack resistance: Not applicable.
Outer conductor integrity: Not applicable.

Cold bend: -40°C ±2°C.

Dimensional stability: +85°C ±2C.

Inner conductor from core: 0.062 inch, maximum.

Inner conductor from jacket: .130 inch, maximum.

Contamination: Not applicable.
Bendability: Not applicable.
Flammability: Applicable.

Flame propagation: Applicable.

Acid gas generation: 2.0 percent, maximum.

Halogen content: 0.2 percent, maximum.

Immersion test:

Tensile strength, percent of unaged minimum: 50.

Elongation, percent of unaged minimum: 50.

Smoke index: 25 maximum.

Toxicity index: 5 maximum.

Durometer hardness (type A) 80 minimum.

Weathering: Applicable.

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Abrasion resistance: 75 cycles minimum (jacket only).

Tear strength: 35 pounds per inch.

Heat distortion: 30 percent maximum.

Physical tests on unaged jacket:

Tensile strength: 1,300 psi, minimum.

Elongation: 160 percent, minimum.

Physical tests on aged jacket:

Air oven:

Tensile strength, percent minimum: 60.

Elongation: percent minimum: 60.

Hot oil immersion:

Tensile strength, percent minimum: 50.

Elongation, percent minimum: 50.

Tensile strength and elongation: 1,300 psi, 160 percent minimum.

Weight: 0.235 Lbs/ft, maximum.

Marking: Applicable.
Marking Durability: 125 cycles (250 strokes), 1 pound.

Workmanship: Applicable

Special requirements:

Magnetic shield continuity: One hundred percent of all finished cable shall be tested for shield continuity prior to shipment. To establish continuity, no more than 25V dc shall be applied across the shield through an appropriate indicator, such as an ohmmeter, light or buzzer.

Magnetic shield: To be applied to the maximum tension possible so as to prevent loosening or creeping but not cause broken ends. Braids shall have no irregularities or loose unwoven strands. There shall be no splices of the completed braid.

Shielding effectiveness:

Frequency range: .00006 - .01 MHz.
Surface transfer impedance, 8.184 milliohms/meter, maximum.

Frequency range: .0101 – 100 MHz.
Surface transfer impedance, 11.296 milliohms/meter, maximum.

Surface transfer impedance measurements performed in accordance with SAE-AS85485. Testing performed with outer conductor shield and magnetic shield connected together on both ends and is considered the “Shield” in the test procedure.

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TABLE II. Qualification and Group A, B, and C Table.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
I	In-process inspection:		4.5	Entire sample		
	Continuity	3.7.1	4.8.2	Entire sample		
	Spark test	3.7.2	4.8.3	Entire sample		
	Voltage withstanding	3.7.3	4.8.4	Entire sample		
	Insulation resistance	3.7.4	4.8.5	Entire sample		
	Visual and mechanical inspection	3.6	4.8.1	Entire sample	Per Table VIII	A
	Diameter measurements.	3.6.1	4.8.1.1	Entire sample	Per Table VIII	A
	Out-of-roundness	N/A				
	Eccentricity of inner conductor	3.6.3	4.8.1.3	2	Per Table VIII	A
	Adhesion of conductors	3.6.4	4.8.1.4	4	Per Table VIII	A
	Marking	3.8	4.8.1.4	Entire sample	Per Table VIII	A
	Marking Durability	3.8.2	4.8.39	3	Per Table VIII	A
	Workmanship	3.10	4.8.1.4	Entire sample	Per Table VIII	A
II	Corona extinction voltage	3.7.5	4.8.6	1	Per Table VIII	B
	Characteristic impedance	3.7.6	4.8.7	1	Per Table VIII	A
	RF transmission loss (attenuation)	3.7.7	4.8.8	2	Per Table VIII	A
	Standing wave ratio (return loss)	N/A				
	Capacitance	3.7.9	4.8.10	1	Per Table VIII	B
	Capacitance stability	N/A				
	Capacitance unbalance	N/A				
	Transmission unbalance	N/A				
	Inductance	N/A				
	Mechanically induced noise voltage	3.7.13	4.8.14	1	Per Table VIII	B
	Time delay	N/A				
	Aging stability	3.7.15	4.8.16	4	Per Table VIII	C

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TABLE II. Qualification and Group A, B, and C Table - Continued

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
II	Stress crack resistance	N/A				
	Outer conductor integrity	N/A				
	Cold bend	3.7.18	4.8.19	3	Per Table VIII	B
	Dimensional stability	3.7.19	4.8.20	1	Per Table VIII	C
	Contamination	N/A				
	Bendability	N/A				
	Flammability	3.7.22	4.8.23	1	Per Table VIII	C
	Flame propagation	3.7.23	4.8.25	1		
	Acid gas generation	3.7.24	4.8.26	1		
	Halogen content	3.7.25	4.8.27	1		
	Immersion tests	3.7.26	4.8.28	1		
	Smoke index	3.7.27	4.8.29	1		
	Toxicity Index	3.7.28	4.8.30	1		
	Durometer hardness	3.7.29	4.8.31	1		
	Weathering	3.7.30	4.8.32	1		
	Abrasion resistance	3.7.31	4.8.33	5		
	Tear strength	3.7.32	4.8.34	6	Per Table VIII	B
	Heat distortion	3.7.33	4.8.35	1	Per Table VIII	C
	Physicals (aged)	3.7.34	4.8.36	1	Per Table VIII	C
	Tensile strength and elongation	3.7.35	4.8.37	1	Per Table VIII	C
	Weight	3.9	4.8.24	1	Per Table VIII	B
	Magnetic shield continuity	See this Specification Sheet	See this Specification Sheet	1	100% inspection	
	Magnetic shield	See this Specification Sheet	See this Specification Sheet	1		
	Surface transfer impedance	See this Specification Sheet	See this Specification Sheet	1		

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Part or Identifying Number (PIN): M17/234-00001

TABLE III. Tools (Installing and Removing).

Bin Code	Inner Contact Tooling		Outer Contact Tooling		Installing Tool	Removal Tool
	Basic Crimping Tool	Contact Positioner	Basic Crimping Tool	Die Part Number		
660	MH992 Daniels	K1605 Daniels	M22520/5-01	M22520/5-03 Cavity A	M81969/8-09 OR M81969/14-04	M81969/8-10 OR M81969/14-04

Amendment notations: The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-17, this document references the following:
SAE-AS85485

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project: 6145-2024-024)

Review activities:
Army - AR, AT, CR4
Navy - AS, MC, OS
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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